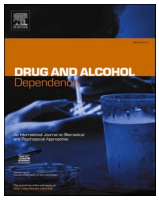
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Short communication



Pilot randomized controlled trial of a tailored personalized normative feedback intervention for college student alcohol use

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ABSTRACT

Purpose: Heavy drinking is common on college campuses, and personalized normative feedback interventions (PFI) are commonly implemented to reduce heavy drinking behaviors. The current study tested the preliminary efficacy of a web-based PFI intervention, Web-BASICS (Brief Alcohol Screening and Intervention for College Students), tailored for participant selection of their normative peer group to reduce descriptive drinking norms and alcohol use compared to a control condition.

Methods: College students ($N = 248$) were recruited to participate in a study on alcohol use. Participants were randomized to receive Tailored Web-BASICS or a control condition and were asked to complete a survey at baseline and three months post-intervention.

Results: Most participants (77 %; $n = 97$) assigned to receive Tailored Web-BASICS chose to have normative feedback tailored by race, ethnicity, gender, sexual orientation, and university. Descriptive drinking norms mediated the effect of Tailored Web-BASICS on alcohol use. Those assigned to Tailored Web-BASICS had reductions in descriptive drinking norms, thereby reducing alcohol use at three months post-intervention relative to control.

Conclusion: Tailored Web-BASICS may be a useful strategy to reduce alcohol use for all college students, irrespective of identity. Future research is needed to assess the efficacy of this approach in a large randomized controlled trial.

Clinical trial.gov registration: NCT06776796

1. Introduction

Heavy drinking is a significant public health problem among college students, with 65 % drinking alcohol monthly (Johnston et al., 2012). Roughly 40 % of students report heavy episodic drinking (HED; 4 drinks for women/5 drinks for men over two hours) in the past month (Wechsler and Nelson, 2008), over 20 % report engaging in HED at least three times in the past two weeks (Hingson et al., 2017), and while alcohol use for men is decreasing, alcohol use among women is on the rise (White, 2020). Rates of HED in the past two weeks differ by gender identity (34 % cisgender women; 36 % cisgender men; 25 % gender minorities), sexual orientation identity (35 % heterosexual, 36 % bisexual, 35 % gay, lesbian, or queer), and racial and ethnic identity (25 % Black, 35 % Latine, and 38 % White; Ranker and Lipson, 2022). College HED is associated with poor academics, interpersonal violence, unintentional injury and deaths, and subsequent alcohol use disorder

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(AUD; Knight et al., 2002; Labouvie, 1996; Peralta et al., 2011; Perkins, 2002b).

The impact of alcohol use is not equitable. Women experience more alcohol-related consequences than men (White, 2020), and Black and Latine populations are disproportionately impacted by alcohol use, with higher rates of AUD and alcohol consequences despite drinking less than their White peers (Delker et al., 2016; Wilson et al., 2016). Further, sexual and gender minority (SGM) college students are at greater risk for alcohol-related consequences compared to their cisgender and heterosexual peers (Fahey et al., 2023; Hughto et al., 2021; Talley et al., 2016; Woodford et al., 2012). Specifically, SGM students had 2.76 % more consequences per drink compared to cisgender heterosexual peers (Schipani-McLaughlin et al., 2022). These disparities are commonly attributed to minority stress resulting from stigma and discrimination (Brem et al., 2024; Dyar et al., 2019; Hendricks and Testa, 2012; Meyer, 2003), in combination with other forms of structural oppression (Mereish, 2024). It is important to develop and evaluate alcohol interventions that are tailored for diverse student populations.

1.1. Web-brief alcohol screening and intervention for college students (BASICS)

Web-BASICS is a personalized normative feedback intervention (PFI) which includes personalized normative feedback, psychoeducation, and motivational interviewing techniques (Larimer et al., 2023). PFIs are the gold standard for addressing alcohol use on college campuses because they are low cost, easily disseminated, and empirically supported (Baer et al., 1991; Cadigan et al., 2015; Dotson et al., 2015; Labrie et al., 2013; NIAAA, 2015; Pueyo-Garrigues et al., 2024; Riper et al., 2009; Saxton et al., 2021). PFIs are based on social norms theory, which suggests individuals are significantly influenced by the behaviors and attitudes viewed as normative within their social groups (Cialdini and Trost, 1998). PFIs work by addressing misperceptions of peers' drinking, or descriptive drinking norms, to correct potential overestimation (Moreira et al., 2012; O'Grady et al., 2011). Descriptive drinking norms strongly predict drinking behavior among college students, with those who overestimate norms consuming alcohol at higher rates than those with accurate perceptions (Perkins, 2002a; Reid and Carey, 2015).

In PFIs, normative feedback reference groups are typically predetermined and based on binary gender and/or the average student at a given university. However, a study using a nationally representative sample found that the use of a "typical" student as the reference group for drinking norms may not be perceived as representative among minoritized students (McCabe et al., 2019). Another study found that for Latine students, ethnicity-specific norms were associated with alcohol use at a university with a large Latine population but not at a university with a small Latine population (LaBrie et al., 2012). Therefore, it is reasonable to propose that PFIs may be less effective for students with marginalized identities, including Black, Latine, or SGM students. Web-BASICS tailoring has not previously been tested for Latine, Black, or SGM students, only for Asian students, finding that when comparing 8 different PFI identity conditions, the typical student was the most parsimonious and effective PFI comparator (Labrie et al., 2013). Representation of Black, Latine, and SGM college students has more than doubled between 2000 and 2019 (The Williams Institute, 2019) and is much higher now than it was in Labrie et al.'s (2013) seminal study over 10 years ago highlighting the importance of assessing Web-BASICS tailoring for current student populations.

1.2. Current study

To address this significant gap, the current pilot study used a randomized controlled trial to compare the efficacy of an adapted Web-BASICS, which tailored normative feedback for alcohol use by choice of identity (Tailored Web-BASICS), to a control condition. We hypothesized that Tailored Web-BASICS would result in decreased alcohol use three months post-intervention across all identities compared to the control condition and that reductions in descriptive drinking norms would mediate this reduction.

2. Methods

2.1. Participants and procedures

College students ($N = 248$; M=20.89 years old, SD=1.54 years old) were recruited using flyers and emails to participate in a study on alcohol use. Participants were eligible if they were full-time college students, 18–25 years old, reported at least one instance of past-month HED, and identified as either Black, Latine, or White and SGM. Due to limited normative data from a random sample of students from the university, individuals who identified as intersex or as both gender diverse and heterosexual were excluded. Most participants (80 %) identified as cisgender women ($n = 199$), and the remaining identified as cisgender men (15 %, $n = 37$) or a gender minority (5 %, $n = 12$). A total of 44 % of participants identified as a sexual minority ($n = 108$). Further, 60 % identified as Black ($n = 150$), 23 % as Latine ($n = 58$), and 16 % as White ($n = 40$).

The study was approved by the university's Institutional Review Board and was registered at clinicaltrials.gov (NCT06776796). Data collection occurred between August 29, 2023 and January 21, 2024. Participants completed a 15-minute baseline survey where they selected their chosen normative feedback group and were randomized to receive either Tailored Web-BASICS ($n = 126$) or information about substance use and mental health in the control group ($n = 122$). Following the three-month post-intervention survey, participants were paid up to $30. See Fig. 1 for CONSORT Diagram.

2.1.1. Tailored Web-BASICS

Web-BASICS is a well-established PFI that pairs motivational interviewing with psychoeducation and personalized normative feedback (Baer et al., 1991; Larimer et al., 2011; Larimer et al., 2023). It targets alcohol use through algorithm-driven personalized normative feedback and a social norms approach (Baer et al., 1991), correcting misperceptions by comparing individual drinking to descriptive (perceived) and actual drinking norms. Tailored Web-BASICS was adapted to allow participants to choose the following normative comparison group(s) on which to receive personalized normative feedback for alcohol use (average drinks per week, average drinking days per week, and average drinks per drinking day): 1) average student at their university; 2) average student of their gender at their university; 3) average student of their sexual orientation at their university; 4) average student of their race or ethnicity at their university; or any combination of options 2, 3, and 4. Actual drinking norms for these groups were obtained using a random sample of students.

2.1.2. Control condition

Participants in this condition were directed to their university's health website to review information on substance use and mental health.

2.2. Measures

2.2.1. Demographics

Demographics assessed included age, gender, sexual orientation, race, and ethnicity.

2.2.2. Alcohol use

Alcohol use was assessed using the Daily Drinking Questionnaire (DDQ; (Collins et al., 1985) at baseline and three months post-intervention. Participants reported the number of standard drinks they consumed each day of a typical week within the past month. A sum score was calculated to reflect the number of drinks per week.

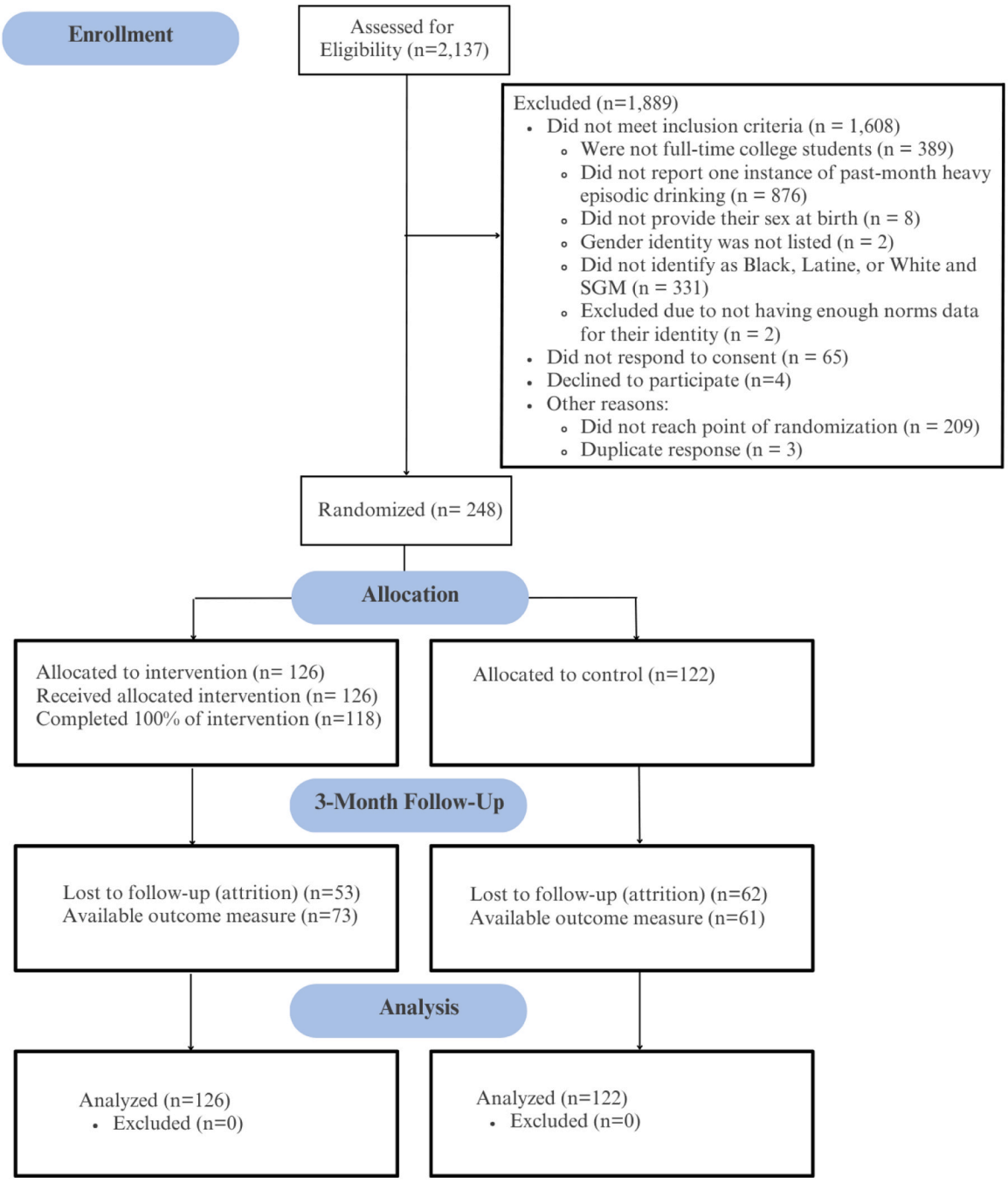


Fig. 1. CONSORT diagram.

2.2.3. *Descriptive drinking norms*

Descriptive drinking norms were assessed using the Drinking Norms Rating Form (DNRF; (Baer et al., 1991) at baseline and at 3 months to evaluate participants' perceptions of peer alcohol use. Participants estimated the typical number of standard drinks consumed each day in a typical week by a student at their university with the same selected identity.

2.3. *Analytic strategy*

MPlus8 was used to test the mediation model examining whether indirect effects of condition on drinks per week three months post-intervention varied by drinking norms (model 4 in Hayes, 2022). Covariates included baseline drinks per week, baseline drinking norms, age, gender, sexual minority status, ethnicity, and race to account for potential confounding. Statistical effects were determined with p-values of less than .05 for direct effects and confidence intervals that do not include zero for indirect effects per guidance by Hayes (2022). Because this was a pilot study, effect sizes were also reported using Cohen's *d* (0.2 =small effect; 0.5 =medium effect; 0.8 =large effect).

3. Results

3.1. *Descriptive statistics*

See Table 1 for baseline and follow-up descriptive statistics for drinks per week on average and descriptive drinking norms. Most participants (77 %; $n = 97$) assigned to receive Tailored Web-BASICS chose

Table 1
Descriptive statistics and mediation analysis.

	Linear Regression with Descriptive Drinking Norms as Outcome (n = 248)					Linear Regression with Drinks per Week as Outcome (n = 248)				
	β	*p*	LLCI	ULCI	*d*	*β*	*p*	LLCI	ULCI	*d*
Condition	**−0.213**	**.003**	**−4.183**	**−1.241**	**0.545**	0.083	.263	−0.551	2.592	0.212
Baseline Drinks per Week	**0.308**	**.002**	**0.117**	**0.423**	**0.789**	**0.292**	**.010**	**0.079**	**0.417**	**0.748**
Baseline Descriptive Drinking Norms	**0.272**	**.005**	**0.091**	**0.354**	**0.696**	−0.024	.778	−0.019	0.094	0.614
3-month Descriptive Drinking Norms	-	-	-	-		**0.468**	**< .001**	**0.277**	**0.630**	**1.198**
Age	0.010	.900	−0.479	0.558	0.026	0.059	.391	−0.220	0.690	0.151
Cisgender woman (vs. cisgender man)	−0.086	.405	−4.114	1.355	0.220	0.123	.189	−0.488	4.310	0.315
Gender minority (vs. cisgender man)	0.057	.568	−3.162	6.532	0.146	−0.014	.838	−3.641	2.834	0.0358
Sexual minority (vs. heterosexual)	−0.022	.808	−2.146	1.594	0.056	−0.067	.372	−2.406	0.731	0.171
Latine (vs. Black)	0.008	.939	−2.561	2.331	0.020	−0.048	.628	−0.696	1.661	0.123
White (vs. Black)	−0.072	.452	−1.134	3.010	0.184	0.100	.244	−1.262	0.479	0.256
Indirect Effects										
Condition on Drinks per Week Through Norms						**−0.100**	**.007**	**−1.961**	**−0.499**	**0.256**

Descriptive Statistics for Outcome Variables

	Full Sample			Tailored Web-BASICS			Control		
	N	*M*	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*
Baseline Drinks per Week	248	9.496	7.273	126	9.024	7.261	122	9.984	7.283
3-month Drinks per Week	145	6.048	5.922	78	5.782	6.567	67	6.358	5.101
Baseline Descriptive Drinking Norms	235	9.970	7.757	121	9.554	7.312	114	10.412	8.206
3-month Descriptive Drinking Norms	143	8.105	6.142	78	6.577	5.799	65	9.938	6.082

normative feedback tailored by all three identities offered. There were no statistically significant differences between participants who chose to have normative feedback tailored by all three identities or less based on race and ethnicity (Latine vs. Black, χ^2=1.08, p = .30; White vs. Black, χ^2=.454, p = .50), sexual orientation (sexual minority vs. heterosexual, χ^2=1.00, p = .32), or gender identity (cisgender woman vs. cisgender man, χ^2=.50, p = .48; gender diverse vs. cisgender man, χ^2=.13, p = .72).

3.2. Mediation analysis

The direct effect of the intervention condition on drinks per week was not significant (β=.083, p = .263; see Table 1). However, with the inclusion of descriptive drinking norms, there was an indirect effect of the intervention condition on drinks per week through changes in descriptive drinking norms (β=-1.00, p = .007; see Table 1). Those in the Tailored Web-BASICS condition compared to control had lower descriptive drinking norms three months post-intervention (β=-2.13, p = .003), and this was significantly associated with drinks per week three months post-intervention (b=0.468, p = .010; see Table 1).

4. Discussion

The current study was a pilot randomized controlled trial examining the preliminary efficacy of Tailored Web-BASICS. Although there was no significant main effect of condition on alcohol use directly, Tailored Web-BASICS was associated with significant reductions in descriptive drinking norms leading to decreased alcohol use three months post-intervention. This is the first study to tailor PFI or Web-BASICS with the inclusion of Black, Latine, and SGM college students. Results underscore the potential utility of integrating participant choice for normative comparison groups into PFIs and Web-BASICS, providing a promising strategy for reducing drinking on college campuses.

Only one study to date has tailored PFI reference groups by race, Greek membership, year in school, and gender; it found that university- and gender-specific norms were most impactful, but it did not examine Black, Latine, or SGM communities (Labrie et al., 2013). Although PFIs, including Web-BASICS, show reductions in alcohol use among college students overall (Cadigan et al., 2015; Dotson et al., 2015; Pueyo-Garrigues et al., 2024; Saxton et al., 2021), their reliance on generalized descriptive drinking norms may limit effectiveness for all college students, whose drinking behaviors are shaped by diverse identity-based contexts.

In this study, most (77 %) participants elected to have feedback tailored, highlighting interest in this approach. Moreover, there were no demographic differences among those who selected full tailoring and those who chose less groups, suggesting students prefer tailoring when given the option, regardless of identity. This is consistent with emerging literature indicating that gender- or university-specific descriptive drinking norms may not be as effective for all students (Labrie et al., 2013; Martin et al., 2013; McCabe et al., 2019). With Black, Latine, and SGM students representing a growing proportion of the college population and facing higher rates of AUD (ACHA, 2020; Cantor et al., 2020; Delker et al., 2016; Wilson et al., 2016), inclusion of these groups in research is increasingly critical to enhancing the relevance and accuracy of feedback, thereby addressing the limitations of current approaches and potentially increasing drinking behavior change. Our findings highlight the importance of incorporating intersectional identities into PFIs to create more effective and inclusive interventions. Further, our study was conducted at a Minority Serving Institution that is also a Predominately Black Institution. Previous research found that for Latine students, ethnicity-specific norms were associated with alcohol use at a university with a large Latine population, but not at a university with a small Latine population (LaBrie et al., 2012). Therefore, these findings may be limited to Minority Serving Institutions where there is a large population of students from minoritized races and ethnicities. Nonetheless, future research should conduct a fully powered randomized controlled trial to assess the efficacy of this approach across multiple university types in combination with implementation research to ensure that implementation barriers and facilitators can be assessed.

Results should be interpreted with several limitations. First, as a pilot trial, this study was not fully powered to detect potential direct effects of Tailored Web-BASICS on changes in alcohol use. Future research should assess efficacy using a large randomized controlled trial. Second, a true mediation analysis was not possible because only two timepoints were collected. Future research should include more time points. Third, no data was collected on how much time, if any, was spent on the website in the control condition. Fourth, outcomes based on intersecting identities were not possible to examine due to not having enough power. However, future research should examine the role of intersecting identities on outcomes. Finally, this was a first step at examining the Tailored Web-BASICS approach. Future research should include diverse identities to examine if this approach is effective for all college students.

5. Conclusion

Tailored Web-BASICS was associated with reduced descriptive drinking norms, which was associated with reduced alcohol use three months post-intervention. Although additional research is required to determine if Tailored Web-BASICS offers improved alcohol use outcomes over traditional PFI approaches, these findings have important implications. Most participants chose to tailor their PFI. When individuals choose their normative comparison group, the feedback may be interpreted as more personally relevant. Tailored feedback provided on these identities resulted in reduced alcohol use even at three months post-intervention, relative to control. These findings underscore the value of this novel approach.

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CRediT authorship contribution statement

Amanda K. Gilmore: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Ruschelle M. Leone:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Katherine Reuben:** Writing – review & editing, Writing – original draft. **Jamani B. Garner:** Writing – review & editing, Writing – original draft. **Nyla Flowers:** Writing – review & editing, Writing – original draft. **Tristin Chipman:** Writing – review & editing, Writing – original draft. **Emily E. Patton:** Writing – review & editing, Writing – original draft. **Julianna Capobianco:** Writing – review & editing, Writing – original draft. **Zoe Y. Zong:** Writing – review & editing, Writing – original draft. **Cynthia A. Stappenbeck:** Writing – review & editing, Writing – original draft. **Jessalynn R. Ellis:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Data curation. **Kennicia Fortson:** Writing – review & editing, Investigation, Formal analysis, Data curation. **K. Nicole Mullican:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Data curation. **Mary E. Larimer:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors have no conflict of interest to declare.

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